

 Digital Egypt Pioneers Initiative (DEPI)

Supply Chain **Insights**

Comprehensive Operational Dashboard & Performance Analytics

 **Instructor:** Eng. Wael Gomaa

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600 × 400

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Project Team (2)

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Executive Summary

Transforming Data into Actionable Strategy

This project presents an interactive **Supply Chain Dashboard** delivering a 360-degree view of core logistics, inventory management, order fulfillment, and shipping performance.

By streamlining complex transactional datasets, the dashboard simplifies operational performance monitoring, pinpoints bottlenecks, and enhances cross-functional decision-making.

600 × 400

Data Foundation & Quality

DEPI Data Source

The dataset was sourced directly from the **Digital Egypt Pioneers Initiative (DEPI)** platform, specifically curated for educational excellence and professional analytics training.

It encompasses granular transactional records covering suppliers, inventory stock levels, order timelines, regional shipments, and sales revenue metrics.

Data Quality Control (QC)

Prior to dashboard development, rigorous **Data Quality Control (QC)** protocols were executed to sanitize and structure the raw dataset.

Key QC operations included handling missing values, standardizing date formats, removing duplicate records, and validating numeric metrics to ensure total accuracy.

⚠ Problem Statement



Inventory Risks

Frequent imbalances between overstocked slow-moving items (tying up working capital) and stockouts on high-demand items.



Logistics Inefficiencies

Inability to track supplier lead time delays and unpredictable regional shipping costs impacting profit margins.



Unclear Profitability

Lack of visibility into low-margin products generating high sales volume versus true profit-driving product categories.

? Key Analytical Questions



Product Performance: Which products drive top-tier profitability versus high-sales, low-margin items?



Supplier Reliability: Which vendors incur the highest delay rates and logistics cost overruns?



Shipping & Transport: What are average delivery durations and regional bottleneck locations?



Inventory Health: What is the turnover efficiency and stockout exposure across warehouse nodes?

Analytical Workflow

1

Data Collection

In-depth ingestion of raw supply chain records from DEPI platform.

2

Data Cleaning

QC scrubbing, null handling, and metric validation.

3

Data Modeling

Building relational Star Schema & dimensional relationships.

4

DAX Measures

Calculating custom KPIs, time intelligence & profit margins.

5

Report Design

Selecting optimized visuals & user UI layout structure.

6

Dashboard

Finalizing interactive filter slices & executive views.

Visualization Methodology



Bar & Column Charts

Used to contrast discrete numerical metrics across suppliers, shipping methods, and product categories.



Pie & Donut Charts

Ideal for displaying regional market share proportions and inventory status distribution at a glance.



Tree Maps & Multi-Series

Hierarchical visualization of multi-level category data alongside grouped comparisons across time periods.

Dashboard Modules

5 Core Analytical Viewports

The final interactive dashboard features dedicated exploration tabs for total operational clarity:

- ✓ Supplier & Logistics Efficiency
- ✓ Customer Demographics & Sales
- ✓ Product Performance & Margins
- ✓ Inventory & Stock Management



Profitability & Supplier Insights

Profitability Insights

Pareto Distribution: A concentrated group of top-performing items drives the majority of net profit (80/20 rule).

Margin Divergence: Certain high-volume items yield surprisingly low profit margins due to elevated holding and freight costs.

Supplier Performance

Lead Time Variances: Identified specific vendors responsible for recurring shipment delays, directly impacting downstream order fulfillment.

Cost Disparities: Significant price variances between suppliers for identical raw materials impact gross margins.

Inventory & Shipping Insights

Inventory Optimization

Working Capital Lockup: Overstocking is localized in slow-moving categories, creating unnecessary holding expenses.

Stockout Vulnerability: Fast-moving items exhibit tight safety buffers, increasing lost-sale risks during demand spikes.

Logistics & Transit

Geographic Bottlenecks: Specific delivery routes suffer consistent delays due to customs and transit infrastructure.

Express vs Cost Balance: Premium shipping options boost customer satisfaction score but require volume consolidation.

Key Value Drivers

80 / 20

Pareto Profit Focus

Focusing resources on key high-margin SKUs.

5

Integrated Dashboards

Unified view across suppliers, inventory & sales.

100%

Data QC Verified

Ensuring reliable data for executive decision making.

 Q & A Session

Thank You!

We welcome your questions and feedback regarding our
Supply Chain Insights project.

DEPI Team 2 | Supervised by Eng. Wael Gomaa

Image Sources



<https://blog.fhysics.net/hubfs/133.png>

Source: blog.fhysics.net